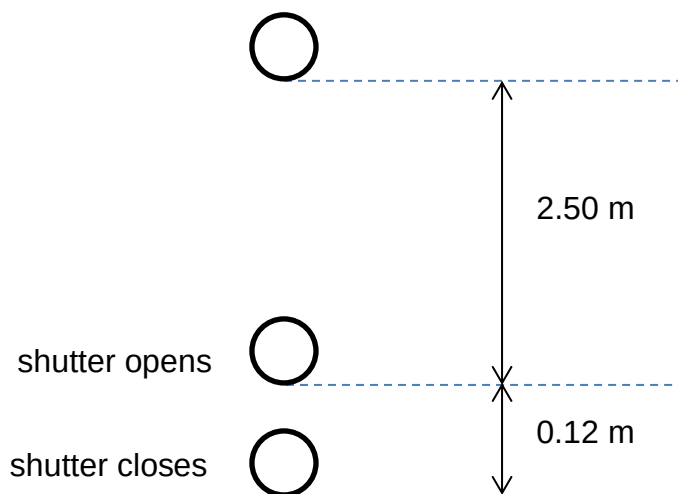


- 2 A photographer wishes to check the time for which the shutter on a camera stays open when a photograph is being taken. It is found that before the shutter opens, the ball falls 2.50 m from rest. During the time that the shutter remains open, the ball falls a further 0.12 m as illustrated.



What is the time that the shutter remains open?

A 0.017 s

B 0.16 s

C 0.71 s

D 0.73 s

A Let time taken to fall 2.50 m be t .

$$s = ut + \frac{1}{2} at^2$$

$$2.5 = 0 + \frac{1}{2} (9.81) t^2$$

$$t = 0.714 \text{ s}$$

Let time taken to fall 2.62 m be t_1 .

$$s = ut + \frac{1}{2} at^2$$

$$2.5 + 0.12 = 0 + \frac{1}{2} (9.81) t_1^2$$

$$t_1 = 0.731 \text{ s}$$

$$\text{The time when the shutter remains open} = t_1 - t = 0.017 \text{ s}$$

3 A projectile is fired at a speed of 90 m s^{-1} at sea level at an angle of $\theta = 31.6^\circ$ with